Homework 2 Solutions IEORE4004, Fall 2025
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Problem 1

Solve

$$\max c^T x - d^T \lambda : \; Ax - \lambda \leq b, \; x \geq 0, \; 0 \leq \lambda \leq k,$$

where $\lambda = (\lambda_1, \ldots, \lambda_m)^T$, $d = (d_1, \ldots, d_m)^T$, $k = (k_1, \ldots, k_m)^T$.

Problem 2

First solve (1) $\max c^T x : x \in P$. If it is infeasible, no solution to the problem exists; if it is unbounded, then the global problem is unbounded. Thus, suppose that the value of the optimal solution is M. Now add $c^T x = M$ as a constraint to P, as to obtain P'. Note that all and only feasible solutions to P' are the optimal solutions to (1). Last, solve $\max d^T x : x \in P'$ to obtain the required solution.

Problem 3

Proof. $\Rightarrow$) If $\sum_{v \in V} d_v = 0$, then the problem has a solution.

We prove it by induction on the number of nodes v with $d_v \neq 0$.
Base case: If $k = 0$, the statement trivially holds since no flow is required.
Inductive step: Assume the statement holds for $1, 2, \ldots, k-1$ nodes. We must have $k > 2$, since the case $k = 1$ is vacuous. Let v and u be nodes where $d_v > 0$ and $d_u < 0$, respectively. Send $\min\{d_v, |d_u|\}$ units of flow from u to v, reducing the number of nodes with nonzero demand by at least one. Let $\bar{x}$ be the flow obtained (which is therefore non-zero on exactly one arc). By the inductive hypothesis, the problem has a solution for $k-1$ nodes. Let $\hat{x}$ be the corresponding flow. Then $\bar{x} + \hat{x}$ can easily be seen as solution to the original problem.

$\Leftarrow$) If the problem has a solution, then $\sum_{v \in V} d_v = 0$.

If $\sum_{v \in V} d_v \neq 0$, total supply and demand are unbalanced, and no feasible flow can satisfy the supplies and the demands. In fact, the flow needs to satisfy

$$\sum_{a=(u,v) \in A} x_a - \sum_{a=(v,u) \in A} x_a = d_v \quad \text{for } v \in V.$$

By summing all such equations, the left-hand side cancels out and we have

$$0 = \sum_{v \in V} d_v \neq 0,$$

thus no such flow exists. Thus, $\sum_{v \in V} d_v = 0$ is a necessary condition. □

Second part: No. Take any instance with $d_v = 2$ but total capacity of arcs going into v strictly less than 2.

Problem 4

Let: x_t be the number of units produced at time t; y_t be the number of units stored between time t and time $t+1$.

Then the Linear Programming model can be formulated as follows to minimize the overall cost:

$$\text{Minimize} \quad \sum_{t=1}^{T} c_t x_t + \sum_{t=1}^{T-1} e_t y_t$$

subject to

$$\begin{aligned} x_t \le u_t, &\quad \forall t = 1, 2, \ldots, T \\ y_t \le q_t, &\quad \forall t = 1, 2, \ldots, T \\ x_t + y_{t-1} = d_t + y_t, &\quad \forall t = 1, 2, \ldots, T \\ x_t, y_t \ge 0, &\quad \forall t = 1, 2, \ldots, T \end{aligned}$$

where $y_0 = 0$ and $y_T = 0$.

From the computation we notice that many variables are set to 0 and, as long as demands and bounds are integer, the optimal solution is integer.

Problem 5

The dual is

$$\begin{array}{lllll} \text{minimize} & 40\,y_1 \;+\; 15\,y_2 \;+\; 30\,y_3 & & & \\ \text{s.t.} & 3y_1 + y_2 + y_3 & \ge & 2 & (\text{since } x_1 \ge 0) \\ & y_1 - y_2 + y_3 & \le & -1 & (\text{since } x_2 \le 0) \\ & y_1 + 2y_2 - y_3 & = & 1 & (\text{since } x_3 \text{ free}) \\ & y_1 \le 0, \quad y_2 \text{ free}, \quad y_3 \ge 0 & & & \end{array}$$

The problems can be solved in many ways – e.g., using a software. The optimal solution is $(40, -45, -35)$. Then we can use complementary slackness to obtain the optimal dual solution $(-3/2, 3, 7/2)$.

Problem 6

The dual of a generic min-cost flow problem on a network $D(V, A)$ in the variables y, z (corresponding, respectively, to flow and capacity constraints) is given as follows

$$\begin{array}{llllll} \max & \sum_{i \in V} d_i y_i & + & \sum_{a \in A} u_a z_a & & \\ & y_i - y_j & + & z_{ji} \;\le\; c & & \text{for all } (j, i) \in A \\ & & & z \;\le\; 0. & & \end{array}$$

When specialized to the instance from the lecture notes, the constraints from above become

$$
\begin{aligned}
y_C - y_D &+ z_{DC} \leq 1 \\
y_H - y_D &+ z_{DH} \leq 6 \\
y_H - y_C &+ z_{CH} \leq 2 \\
y_N - y_C &+ z_{CN} \leq 3 \\
y_M - y_N &+ z_{NM} \leq 5 \\
y_M - y_H &+ z_{HM} \leq 4 \\
& \quad z \leq 0.
\end{aligned}
$$